



Kenya Certificate of Secondary Education (KCSE)

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CHEMISTRY**Paper 3****(PRACTICAL)****MARKING SCHEME**

1.

Table II	1	2	3
Final burette readings(cm^3)	11.4	22.8	34.2
Initial burette readings(cm^3)	0.0	11.4	2.8
Volume of HCl used cm^3 (solution N)	11.4	11.4	11.4

(3 mks)

CT - 1 mk

Penalize $\frac{1}{2}$ mk for wrong arithmetic, unrealistic figures, incomplete table to a maximum of $\frac{1}{2}$ mk

- If only one experiment done = 0
- If 2 experiment done = $\frac{1}{2}$ mk
- If all experiments done = 1 mk

DP – $\frac{1}{2}$

1 or 2 dp used consistently

Penalize fully if mixed or missing

ACC. = $\frac{1}{2}$ mk ± 0.1 deviation from S.V $\frac{1}{2}$ mk

Otherwise penalise fully.

P.A = 1 mk

Values average within ± 0.2 and correct working shown and correct answer given. 1 mkIf no working shown but correct answer $\frac{1}{2}$ mkIf wrong units penalize $\frac{1}{2}$ mk

If no units ignore.

F.A = 1 mk

 ± 0.1 deviation from school average titre 1 mk ± 0.2 deviation from school average titre $\frac{1}{2}$ mk

If wrong units penalize fully if no units ignore.

a) (i)
$$\frac{11.4 + 11.4 + 11.4}{3} = 11.4$$

ii)
$$\frac{0.2 \times 25}{1000} = 0.005 \text{ moles}$$



iii) Mole ratio of HCl : NaOH is 1:1

$$\frac{0.005 \text{ moles} \times 1000}{11.4} = 0.4386 \text{ mol/l or mol/dm}^3 \text{ or } 0.4386 \text{ M}$$

TABLE II

Marked as table I

	1	2	3
Final burette readings(cm^3)	40.0	40.0	40.0
Initial burette readings(cm^3)	0.0	0.0	0.0
Volume of solution L used cm^3	40.0	40.0	40.0

(4mks)

CT – 1

DP – $\frac{1}{2}$

AC – $\frac{1}{2}$

PA – 1

FA – 1

b. (i) $= \frac{40.0 + 40.0 + 40.0}{3} = 40.0 \text{ cm}^3$

(ii) Mole ratio 1:1

Moles in 25 cm^3 of NaOH = moles in 40 cm^3 of HCL
 $= 0.005 \text{ moles}$ as in a (ii) above

(iii) Moles of H1 100 cm^3 of solution L.

$$= \frac{100 \times 0.005}{40} = 0.0125 \text{ moles.}$$

(iv) 1000 cm^3 of solution N = 0.4386 moles

100 cm^3 of solution N:

$$= \frac{0.4386 \times 100}{1000}$$

$$= 0.04386 \text{ moles}$$

(v) Moles of HCl reacted with solid +

Answer in (iv) – answer in (iii)

$$= 0.04386 - 0.0125$$

$$= 0.03136 \text{ moles}$$

(vi) Moles of F_2CO_3 reacted

= Mole ratio 1:2

= F_2CO_2 : HCl

$$= \frac{0.03136}{2}$$



$$= 0.01568 \text{ moles}$$

(vii) RMM = 1g contain 0.03136 moles

1 mole will contain

$$= \frac{1 \times 1}{0.03136}$$

$$= 63.7755 \cong 64$$

$$\text{RFM of F}_2\text{CO}_3 = (64 \times 2 + 12 + 48) = 128$$

$$F = \frac{128 - 60}{2}$$

$$F = 38$$

2. **C-T – 2 mks** as follows

8 readings – 2mk

6 – 7 readings – 1 mk

4 -5 readings – $\frac{1}{2}$ mk

0-4 readings – 0 mk

Readings between (40 and 90s)

NB: Correct working on $1/t$

DP – 2 mks

Time 1 or whole numbers

$1/t$ minimum of 4 dp unless it divides fully.

AC – 1 mk

Tied to 1st reading ± 0.5 deviation from school value.

Trend 1 mk

Time increasing with increase in temperature otherwise penalize fully.

Graphs

a) i) Time(s) x axis labelled correctly with units

L = $\frac{1}{2}$ mk

S = $\frac{1}{2}$ mk

Plot = 1 mk

Curve = 1 mk

NB: Label axis labelled correctly with units or penalize fully

If interchange penalize fully

Scale

Atleast $\frac{1}{2}$ of the grid provided should be occupied.

Plot

8 plots – 1 mk

5 – 7 plots – $\frac{1}{2}$ mk



0 – 4 plots – 0 mk

Curve

A curve descending from left to right. Otherwise penalize fully.

2. a) ii) Plot a graph of $1/t$ against temperature change

L - $\frac{1}{2}$ mk

S - $\frac{1}{2}$ mk

P - 1 mk

C - 1 mk

NB: Mark as a (i) above

- b) As temperature increases rate of reaction also increases

- c) Read $1/t$ at 58°C from candidates graph

e.g. 0.17

$$1/t = 0.17, \quad t = 1/0.17 = 5.8823 \cong 6 \text{ sec}$$

$\frac{1}{2}$ mk showing, $\frac{1}{2}$ mk correct as

- d) Student should draw a tangent at 4.3°C and work out as shown

$$\frac{\Delta \text{Temp}}{\Delta \text{Time}} \quad \text{Or} \quad \frac{\Delta y \text{ axis}}{\Delta \text{in } x \text{ axis}}$$

$$= \frac{56 - 22}{21 - 7} = 2.4286$$

Showing $\frac{1}{2}$ mk

Correct answer $\frac{1}{2}$ mk

3. (a)

Observations	Inference
Efferescence / bubbles / fizzles / colourless gas produced White ppt or glass rod	CO_3^{2-} or HCO_3^- present
White ppt solution \warming but reappears on cooling	Cl- Present
No white ppt formed	Zn^{2+} , Al^{3+} Pb^{2+} Absent

- (b)

Observations	Inference
PH – 4 – 6	-Weakly acidic substance
(i) Indicator paper turns Yellow orange	-R – COOH / H^+ / H_3O^+
(ii) Effervescence / bubbles /	$\text{H}^+_{(\text{aq})}$ ions / H_2O^+ / R-COO



fizzing / colourless gas produced or evolved	
Purple colour persists	R-OH , $\text{>C}=\text{C}<$ — $\text{C}\equiv\text{C}$ — Absent
Sweet smell	R-COOH confirmed present

Conditions for Q3

Reject words instead of chemical symbol

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CHEMISTRY

PAPER 3

CONFIDENTIAL

1. 180cm³ of solution N HCl in a beaker.
2. 180cm³ of solution M 0.2M NaOH in a beaker.
- 3.. 50cm³ of solution D, 2M HCl in a beaker.
4. 100cm³ of solution B, 0.1M Na₂S₂O₃ in a beaker.
5. 10cm³ of 1M HNO₃ in a boiling tube.
6. Ethanol in a stopped container.
7. 5cm³ of conc H₂SO₄ in a test tubewith a dropper.
8. Exact 1g of solid X which is F₂CO₃
9. About 1g of solid L in a dry stoppered container
10. About 1g of solid Q in a stoppered container access to:
11. Methy orange in a dropper
12. 0.5M Ca(OH)₂ in a dropper.
13. 1M Pb(NO₃)₂ in a dropper.
14. 2Ml NaOH solution in a dropper.



15. Distilled water in a wash bottle.
16. Acidified KMnO_4 in a dropper.
17. About 0.5g of Na_2CO_3 per student.
18. Pipette (25cm^3)
19. Burette.
20. Pipette filler.
21. 3 conical flasks. (250cm^3)
22. Stand and clamp.
23. White tile.
24. 100cm^3 glass beaker.
25. Thermometer (-10 to 110°C)
26. 10cm^3 measuring cylinder.
27. 100cm^3 measuring cylinder.
28. Stop watch/ clock.
29. Plain white paper.
30. 2 boiling tube.
31. 6 test tubes in a test tube rack.
32. A glass rod.
33. Metallic spatula.
34. source of heat.

N/B

1. solid X, Solid L and solid Q to be provided by the examining authority.

Preparation of solutions

Solution N is prepared by dissolving 68.8ml of concentrated Hydrochloric acid in 500ml of distilled then top up to 1litre of solution.

2. 1M HNO_3 is prepared by 66ml of conc. HNO_3 in 500ml of distilled water then top up to make 1litre of solution
3. Acidified KMnO_4 is prepared by dissolving 3.16g of KMnO_4 in 400cm^3 of 2M H_2SO_4 then topped up to 1litre of solution by distilled H_2O .
4. 2M H_2SO_4 prepared by dissolving 110cm^3 of conc. H_2SO_4 in 500ml of distilled H_2O then top to 1litre of solution.